MATH MA'AM ACADEMY

Where Every Subject Clicks.

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UNIT 2: TYPES OF MOTION

Expanded revision notes based on the supplied textbook pages, with additional concepts, examples, graph interpretation and useful facts for application-based preparation.

1. WHAT IS MOTION?

Motion is the change in position of an object with respect to a reference point or its surroundings over time. If an object's position does not change with respect to the chosen reference point, it is at rest relative to that reference point.

- Motion and rest are relative. A passenger sitting in a moving train is at rest relative to another passenger, but is moving relative to a person standing on the platform.
- The same object can be at rest with respect to one observer and in motion with respect to another.
- To describe motion, identify the object, the reference point and how its position changes with time.

Examples of relative motion

- A person sitting in a moving bus is at rest relative to the bus but moving relative to the road.
- Trees beside a moving train appear to move backwards to passengers, although the trees are nearly at rest relative to the ground.
- Two vehicles moving in the same direction at the same speed have no change in their separation; each is at rest relative to the other.

2. TYPES OF MOTION

Type	Meaning	Examples
Linear / rectilinear	Motion along a straight line.	Lift moving vertically; car on a straight road; a stone falling vertically (approximately).
Circular	Motion along a circular path around a fixed centre or axis.	Fan blade, hands of a clock, merry-go-round, a stone whirled on a string.
Rotational	A body spins about its own axis.	Earth spinning about its axis; rotating wheel; ceiling fan.
Periodic	Motion or an event repeats itself after equal intervals of time.	Oscillation of a pendulum; Earth revolving around the Sun; hands of a clock.
Oscillatory	To-and-fro motion about a mean or equilibrium position.	Simple pendulum, swing, vibrating tuning fork.
Random	Motion with no fixed path or predictable direction; direction and/or speed changes irregularly.	Flying mosquito, movement of dust particles in air, fish moving irregularly.
Curvilinear	Motion along a curved path.	A thrown ball following a curved trajectory; a car taking a bend.

Important distinctions

- Circular motion describes the path; rotational motion describes spinning about an axis. A point on a rotating fan blade follows a circular path.
- Periodic motion repeats after equal time intervals. Oscillatory motion is a to-and-fro type of motion and is often periodic, but not every periodic motion is oscillatory.
- A moving object can show more than one type of motion. A rolling wheel moves forward while rotating about its axle.

3. DISTANCE AND DISPLACEMENT

Distance	Displacement
Total length of the actual path travelled.	Shortest straight-line change in position from start to finish, with direction.
Scalar quantity: magnitude only.	Vector quantity: magnitude and direction.
Always non-negative; it is zero only if no path is travelled.	Can be positive, negative or zero in one-dimensional motion, depending on chosen direction.
Depends on the actual route taken.	Depends only on initial and final positions.
Never less than the magnitude of displacement.	Magnitude never exceeds distance.

Key relation: Distance ≥ magnitude of displacement. Equality occurs when motion is along a straight line without changing direction.

Examples

- A student walks 5 m east and then 3 m west. Distance = 5 + 3 = 8 m. Displacement = 2 m east.

- A runner completes one full lap of a circular track and returns to the starting point. Distance is the track's circumference, but displacement is zero.
- If a person walks 4 m north and 3 m east, distance = 7 m. The magnitude of displacement = $\sqrt{(4^2 + 3^2)} = 5$ m.

4. UNIFORM AND NON-UNIFORM MOTION

Uniform motion	Non-uniform motion
Covers equal distances in equal intervals of time.	Covers unequal distances in equal intervals, or equal distances in unequal intervals.
Speed remains constant (for straight-line uniform motion, velocity is constant).	Speed or velocity changes with time.
Distance–time graph is a straight line with constant slope.	Distance–time graph is generally curved or has changing slope.
Example: a vehicle travelling on a straight road at constant speed.	Example: a bus moving through city traffic.

A body can move uniformly along a circular path at constant speed, but its velocity changes continuously because its direction changes.

5. SPEED AND VELOCITY

Speed	Velocity
Distance travelled per unit time.	Displacement per unit time.
Scalar quantity; has magnitude only.	Vector quantity; has magnitude and direction.
Average speed = total distance ÷ total time.	Average velocity = total displacement ÷ total time.
Cannot be negative.	Can be positive, negative or zero in one-dimensional motion.
SI unit: metre per second (m/s).	SI unit: metre per second (m/s).

Important formulae

Average speed = Total distance travelled / Total time taken

Average velocity = Total displacement / Total time taken

Speed = Distance / Time (for uniform motion).

- SI unit of distance and displacement: metre (m). SI unit of time: second (s).
- Other common speed units: km/h and cm/s. To convert km/h to m/s, multiply by 5/18. To convert m/s to km/h, multiply by 18/5.
- Speed is the magnitude of velocity at an instant. Average speed is always greater than or equal to the magnitude of average velocity.

Worked examples

- Example 1: A car travels 120 km in 2 hours. Average speed = 120 ÷ 2 = 60 km/h.
- Example 2: A student walks 100 m east in 20 s. Average velocity = 100 ÷ 20 = 5 m/s east.
- Example 3: A runner completes a 400 m lap in 80 s. Average speed = 400 ÷ 80 = 5 m/s; average velocity = 0 because the final position equals the initial position.
- Example 4: Convert 72 km/h into m/s: 72 × 5/18 = 20 m/s.

6. DISTANCE–TIME GRAPHS

- Time is generally plotted on the horizontal (X) axis and distance on the vertical (Y) axis.
- The slope of a distance–time graph represents speed.
- A straight sloping line indicates constant speed (uniform motion). A steeper line represents greater speed.
- A horizontal line indicates that distance from the reference point is not changing; the object is at rest.
- A curved line indicates changing speed (non-uniform motion).
- A distance–time graph cannot slope downwards if distance travelled is measured cumulatively, because total distance cannot decrease.

Reading graph slopes

- If an object covers 20 m in 4 s, the slope/speed is 20 ÷ 4 = 5 m/s.
- If two straight lines are drawn on the same distance–time graph, the steeper line represents the faster object.
- A horizontal line followed by a sloping line means the object was initially at rest and then started moving.

7. CIRCULAR, ROTATIONAL AND OSCILLATORY MOTION

- In uniform circular motion, speed may be constant but velocity changes because direction changes at every point.
- One complete revolution brings a rotating object back to its initial angular position.
- The time taken for one complete oscillation or cycle is called the time period (T). Its SI unit is second.
- The number of complete oscillations per second is called frequency (f). Its SI unit is hertz (Hz).
- Frequency = 1 / Time period; Time period = 1 / Frequency.
- A pendulum's one oscillation is a complete to-and-fro movement, returning to the same position in the same direction.

8. COMMON EXAMPLES AND THEIR MOTION

Object / activity	Type of motion
Drum of a washing machine	Rotational; clothes also move irregularly inside.
Needle of a sewing machine	To-and-fro / reciprocating (periodic when operating regularly).
Kite flying in the sky	Often random or irregular; may also move along a curved path.
Housefly	Random motion.
Potter's wheel	Rotational motion.
Ceiling fan	Rotational; points on blades move in circular paths.
Lift moving between floors	Linear / rectilinear motion.
Pendulum of a clock	Oscillatory and periodic motion.
Earth spinning on its axis	Rotational and periodic motion.
Earth revolving around the Sun	Revolution along an approximately elliptical orbit; periodic motion.

9. ADDITIONAL IMPORTANT CONCEPTS

Scalar and vector quantities

- Scalar: described by magnitude only. Examples: distance, speed, time, mass and temperature.
- Vector: requires magnitude and direction. Examples: displacement, velocity, force and acceleration.

Acceleration

Acceleration is the rate of change of velocity with time. SI unit: m/s^2.

- An object accelerates if its speed changes, its direction changes, or both change.
- A car moving around a circular track at constant speed is accelerating because its direction changes.
- If velocity remains constant, acceleration is zero.

Rest, motion and reference points

- A reference point is a fixed or chosen point used to describe the position and motion of an object.
- There is no absolute description of rest or motion without specifying a reference frame.
- For everyday problems, the ground or road is often chosen as the reference frame.

10. COMMON MISCONCEPTIONS TO AVOID

- Distance and displacement are not the same. Distance depends on the route; displacement depends on start and end positions.
- An object can have zero displacement but non-zero distance after completing a round trip.
- Constant speed does not always mean constant velocity. Direction matters.
- Circular motion at constant speed involves changing velocity.
- A straight distance–time graph means uniform speed, not necessarily that the object is stationary.
- The slope of a distance–time graph gives speed; the slope of a velocity–time graph gives acceleration.
- A body may show combined motions, such as a rolling wheel (translation plus rotation).

11. QUICK FORMULA AND UNIT SHEET

Quantity	Formula / relation	SI unit
Speed	Distance ÷ time	m/s
Average speed	Total distance ÷ total time	m/s
Velocity	Displacement ÷ time	m/s
Average velocity	Total displacement ÷ total time	m/s
Acceleration	Change in velocity ÷ time taken	m/s^2
Frequency	1 ÷ time period	Hz
Time period	1 ÷ frequency	s

12. EXAM APPROACH

- For distance–displacement questions, draw the route and mark the starting and ending points.
- For speed questions, convert all quantities into compatible units before calculating.
- For motion classification, identify the path, whether the object spins, whether it repeats, and whether its direction is irregular.
- For graph questions, inspect the axes and compare slopes rather than judging only by the line's visual length.

Revise the definitions, formulae, graph interpretation and everyday examples. Homi Bhabha questions often combine more than one concept in a single situation.